General phenological model to characterise the timing of flowering and veraison of *Vitis vinifera* L.

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Abstract

Background and Aims: Phenological models, which are based on responses of the plant to temperature, are useful tools to predict grapevine (*Vitis vinifera* L.) phenology in various climate conditions. This study aimed to develop a single process-based phenological model at the species level to predict two important stages of development for *V. vinifera* L.: flowering and veraison.

Methods and Results: Three different phenological models were tested and the model that gave the best results was optimised for its parameters. The chosen model Spring Warming was found optimal with regard to the trade-off between parsimony of input parameters and efficiency. The base temperature (*Tb*) of 0°C calculated from the 60th day (t_0) of the year (for the Northern hemisphere) was found to be the most optimal parameter set tested. This model henceforth referred to as the Grapevine Flowering Veraison model (GFV) was successfully validated at the varietal level using an independent dataset.

Conclusions: A general phenological model, GFV, has been successfully developed to characterise the timing of flowering and veraison for the grapevine.

Significance of the Study: The model is simple for the user, can be successfully applied to many varieties and can be used as an easy predictor of phenology for different varieties under climate change scenarios.

Keywords: *flowering, modelling, phenology, temperature, veraison*

Introduction

Several studies have highlighted the impact of climate change on the timing of seasonal activities in plants and animals such as the advancement of the phenological stages of flowering and leaf unfolding (Menzel and Fabian 1999, Menzel 2003, Cleland et al. 2007). As with other plants, temperature and photoperiod are considered to be fundamental in influencing grapevine (*Vitis vinifera* L.) phenological development and ripening (Winkler et al. 1962, Huglin 1978, Jones and Davis 2000, Jones 2003, Jones et al. 2005, Van Leeuwen et al. 2008, Duchêne et al. 2010). Forecasted increases in temperatures are predicted to cause earlier development and therefore a general advancement of grapevine phenological stages (Duchêne and Schneider 2005, Jones et al. 2005, Webb et al. 2007, Petrie and Sadras 2008, Duchêne et al. 2010). The overall effect predicted for climate change is a shortening of the growth season with maturation occurring during hotter periods of the year (Webb et al. 2007, Hall and Jones 2009, Duchêne et al. 2010).

When the vegetative and reproductive development of the grapevine are well adapted to the local conditions, the grapes at harvest may correspond to a desired combination of sugar, acidity, aromatic and phenolic profile or other desired qualities for the production of high-quality wine (Jones and Davis 2000, Jones et al. 2005, Jones 2006, Van Leeuwen et al. 2008). Grapes which ripen in the warmest part of the summer contain less aromas or aroma precursors (Van Leeuwen and Seguin 2006). As a result of the predicted climate changes, it is possible that varieties currently planted under certain climate conditions today may no longer be adapted to reach maturity under the same conditions in the future. Therefore, understanding how temperature influences the timing of *V. vinifera* L. vegetative and reproductive development as well as identifying varietal specific differences in phenology and maturity is crucial.

Up until now, process-based phenological models for the grapevine work on the assumption that phenological development is mainly regulated by temperature (Williams et al. 1985a,b; Villaseca et al. 1986, Moncur et al. 1989, Riou 1994, Oliveira 1998, Jones 2003, Van Leeuwen et al. 2008; García de Cortázar-Atauri et al. 2009; Caffarra and Eccel 2010, Duchêne et al. 2010, Nendel 2010). These models are driven by a temperature summation from a defined date and above a minimum temperature (threshold) until the appearance of a phenological stage (often judged at 50% level of appearance). Classically, the Spring Warming (SW) model (also known as Growing Degrees Days (GDD)) is the simplest model used to estimate grapevine phenology (bud break, flowering and veraison). This model calculates a summation of daily heat requirements calibrated from a base temperature (usually 10°C for grapevine) and from

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a given date. The sum of the resulting values gives a measure of the state of forcing (heat requirements) in degrees days (°C.d). More complex phenological models also take into account chill requirements (in addition to heat requirements) necessary to break dormancy (for a review see Chuine et al. 2003). Such models have been successfully adapted for other species (Bidabe 1965a,b; Chuine 2000, Cesaraccio et al. 2004, Crepinsek et al. 2006). Recently, phenological models have been successfully developed for the bud break stage of the grapevine, which incorporate both chill and heat units (García de Cortázar-Atauri et al. 2009; Caffarra and Eccel 2010).

Process-based modelling techniques aim to achieve temporal and spatial robustness; i.e. the difference between years and regions in the date of a phenological event for a variety can be explained solely by the differences in abiotic factors such as temperature and photoperiod (Chuine et al. 2003). Such techniques have not been fully explored for flowering and veraison of grapevines. Furthermore, the parameter estimates defining existing indices have not been considered with respect to current modelling capabilities and the availability of new databases. For example, the relevance of the base temperature parameter of 10°C, which was defined by Winkler et al. (1962) for grapevines in California, and often applied in other vineyards and regions has been questioned in prior studies (Moncur et al. 1989, Oliveira 1998, Duchêne et al. 2010) and has not been tested on larger databases (covering more regions and time periods) using more modern and efficient optimisation algorithms for model parameterisation.

The aim of this study was to develop a simple process-based phenological model to predict two important stages of development of *V. vinifera* L., namely flowering and veraison. The best model was selected in terms of efficiency relative to its complexity. The developed model aimed to convey temporal and spatial robustness for the species *V. vinifera* L. By using this approach, all varieties can be compared within the same modelling framework and the model can be applied across different locations. The model can be parameterised further to describe the timing of flowering and veraison for individual varieties.

Materials and methods

Phenological data

Historical data for 50% flowering and 50% veraison were collected from scientific research institutes, extension services ('Chambres d'Agriculture') and private companies in France, Italy, Switzerland and Greece. Dates of 50% flowering were defined as the date when 50% of flowers reached the stage of anthesis, identified as stage 23 on the modified Eichorn and Lorenz (E-L) scale (Coombe 1995). Dates of 50% veraison corresponded to the onset of the ripening period identified as the date when 50% of berries softened or changed from green to translucent for white varieties, or a change of colour of 50% of berries for red varieties (stage 35 on the modified E-L scale).

The phenological observations collected for this study spanned from 1960 to 2007, from 123 different locations (predominantly in France). The observations corresponded to 81 varieties, 2278 flowering observations and 2088 veraison observations (Table 1). Although spanning 55 years, most data collected corresponded to the last 10 years for both phenological stages (Figure 1). Geographically, the data covered 12 of the principal viticultural regions of France, Changins in Switzerland, Veneto and Tuscany in Italy, and the Peloponnese region in Greece. These observations were not collected all at the same time. The first set of observations collected was used to parameterise the models. After model selection, a second set of observations was collected to validate the models. The validation data used corresponded to 11 varieties, 424 observations for veraison and 440 observations for flowering.

Temperature data

Daily minimum and maximum temperatures were collected from meteorological stations situated not more than 5 km away and within ±100 m in altitude for each phenological data site. The average daily temperature was calculated as the arithmetic mean of the daily maximum and minimum temperature.

Phenological models

Three different process-based models were tested: (i) SW with two different forms, (ii) UniFORC and (iii) UniCHILL (Chuine 2000, Chuine et al. 2003).

Spring Warming and UniFORC consider only the action of forcing temperatures. These models assume a phenological stage occurs where t_s corresponds to the day when a critical state of forcing S_f, denoted F^* has been reached (Eqn 1):

$$S_f(t_s) = \sum_{t_0}^{t_s} R_f(x_t) \geq F^* \quad (1)$$

The state of forcing is described as a daily sum of the rate of forcing, R_f, which starts at t_0 (day of the year, DOY) and x_t is the daily mean temperature. The rate of forcing in the SW model is defined by Eqn 2:

$$R_f(x_t) = GDD(x_t) = \begin{cases} 0 & \text{if } x_t < Tb \\ x_t - Tb & \text{if } x_t \geq Tb \end{cases} \quad (2)$$

The SW model contains three parameters, t_0, Tb and F^*, where Tb corresponds to a base temperature above which the thermal summation is calculated.

First, the t_0 value was forced as 1 January, and Tb and F^* were fitted parameters. This corresponds to what is normally considered the GDD model for grapevine except the base temperature was not forced *a priori* to 10°C but was fitted to the data.

Second, t_0 was left unforced; therefore, t_0, Tb and F^* were fitted parameters.

Rate of forcing in the UniFORC model is defined by Eqn 3:

$$R_f(x_t) = \begin{cases} 0 & \text{if } x_t < 0 \\ \dfrac{1}{1 + e^{d(x_t - e)}} & \text{if } x_t \geq 0 \end{cases} \quad (3)$$

The UniFORC model contains four parameters where t_0 is forced as 1 January and d, e and F^* are fitted, with $d < 0$ and $e > 0$.

The third model that was tested, UniCHILL, considers in addition to the UniFORC model, the action of chilling temperatures involved during the dormancy period. It assumes a critical state of chilling (S_c) C^* (Eqn 4) must be reached to break dormancy (t_d), with the rate of development of chilling (R_c) defined by Eqn 5. At this point, a sum of forcing units can start to accumulate until this reaches a critical state F^* as described by Eqn 3. The UniCHILL model has seven fitted parameters a, b, c, C^*, pertaining to the chilling function (Eqn 5) and d, e, F^*,

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Table 1. Quantity and description of data by country, region, number of sites, years and varieties for flowering and veraison observations collected in the database.

Country	Region	Number of sites	Flowering		Veraison		Varieties
			Years	*n*	Years	*n*	
France	Alsace	1	1976–1986 1988–2001	156	1976–1986 1988–2001	154	Cabernet-Sauvignon, Chardonnay, Chasselas,Grenache noir, Merlot, Pinot noir, Riesling, Syrah, Ugni blanc
	Champagne	3	1998–2005	38	1998–2004	34	Chardonnay, Pinot noir
	Loire Valley	15	1981–2007	338	1982–1990 1992–2000 2002–2007	178	Cabernet franc, Cabernet-Sauvignon, Chardonnay, Chasselas, Chenin, Cot, Gamay, Grenache noir, Groulleau, Pinot noir, Riesling, Sauvignon blanc, Syrah, Ugni blanc
	Burgundy	7	1994–2006	105	2004–2005	2	Chardonnay, Gamay, Pinot noir
	Beaujolais	4	1992–2007	45	1992–2007	43	Gamay
	Rhone	1	2000–2007	8	2000–2007	8	Syrah
	Languedoc-Roussillon	27	1960 1962–1966 1979–1982 1986–2007	695	1962–1964 1976–1985 1987–2007	729	Altesse, Arinarnoa, Bourboulenc, Cabernet franc, Cabernet-Sauvignon, Caladoc, Carignan noir, Chardonnay, Chasselas, Chenanson, Chenin, Cinsaut, Colombard, Cot, Egiodola, Ekigaina, Gamay, Gewürztraminer, Gouais blanc, Grenache noir, Jacquère, Marsalan, Marsanne, Merlot, Mondeuse, Mourvèdre, Muscadelle, Muscat blanc à pétits grains, Muscat of Alexandria, Nebbiolo, Petit verdot, Pinotage, Pinot gris, Pinot noir, Pinot meunier, Piquepoul blanc, Piquepoul noir, Poulsard Portan, Riesling, Roussanne, Sangiovese, Sauvignon blanc, Savagnin, Segalin, Sémillon, Syrah, Tannat, Tempranillo, Trousseau, Ugni blanc, Vermentino, Viognier, Xinomavro, Zweigelt blau
	Bordeaux	9	1974–2007	126	1961–1971 1973–2007	215	Cabernet franc, Cabernet-Sauvignon, Merlot, Muscadelle, Sauvignon blanc, Sémillon
	Mid-Pyrénées	4	1999–2004	22	1999–2004	22	Colombard
	Provence-Alpes-Cote d'Azur	45	1976–1977 1979–1980 1982 1984–1986 1994 1997–2007	162	1997–2007	88	Alicante bouschet, Carignan noir, Chason, Cinsaut, Grenache blanc, Grenache noir, Merlot, Mourvèdre, Muscat of Hamburg, Syrah
	Corsica	1	—	—	1998–2006	27	Sangiovese, Sciaccarello, Vermentino
Greece	Peloponnese	3	1997–1998	6	1997–1998	6	Agiorgitiko
Switzerland	Changins	1	1991–1999	337	1991–1999	338	Aligoté, Amigne, Arvine, Bondola, Cabernet franc, Cabernet-Sauvignon, Chardonnay, Charmont, Chasselas, Cornalin, Diolmoir, Doral, Gamaret, Garanoir, Gewürztraminer, Grenache noir, Humagne blanc, Humagne rouge, Kerner, Marsanne, Merlot, Mourvèdre, Muscat blanc à pétits grains, Nebbiolo, Pinot blanc, Pinot cortaillod, Pinot gris, Pinot mariafield, Rauschling, Riesling, Sangiovese, Sauvignon blanc, Savagnin, Sémillon, Sylvaner, Syrah, Ugni blanc
Italy	Veneto Tuscany	2	1964–1981 1985–2006	240	1964–1981 1984–2006	244	Arinarnoa, Cabernet franc, Cabernet-Sauvignon, Chardonnay, Merlot, Petit verdot, Pinot noir, Sangiovese, Sauvignon blanc, Ugni blanc
Total		123		2278		2088	81 varieties

Not all varieties are present for both flowering and veraison within a given region; new crosses (unnamed) are not listed under varieties (although the data is included in the observations, site and years). *n* corresponds to the total number of observations for each phenological stage within a region.

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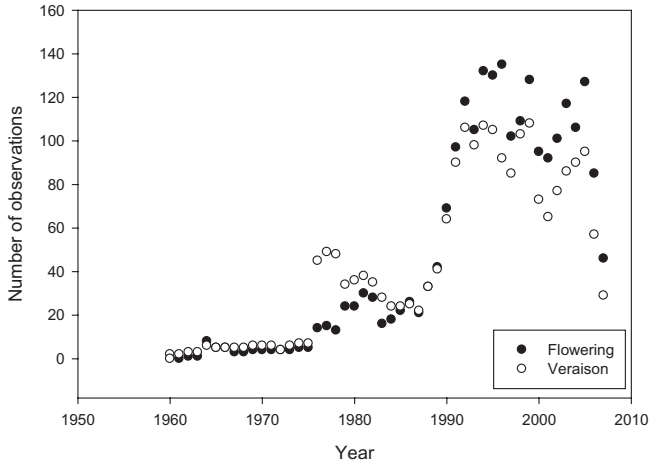


Figure 1. Distribution of phenology data by year for the complete phenological database of flowering and veraison. Closed circles (●) represent flowering data; open circles (○) represent veraison data.

pertaining to the forcing function. t_0 is fixed at 1 September for S_c such that

$$S_c(t_d) = \sum_{t_0}^{t_d} R_c(x_t) \geq C^* \quad (4)$$

and

$$R_c(x_t) = \frac{1}{1 + e^{a(x_t - c)^2 + b(x_t - c)}} \quad (5)$$

Model parameterisation and selection

All data for flowering and veraison across all varieties from the parameterisation dataset were used to fit the most accurate model of the timing of flowering and veraison at the species level. Model parameters were fitted using the simulated annealing algorithm of Metropolis following Chuine et al. (1998). The best model was selected based on three criteria: (i) the model with the highest efficiency, i.e. that gives the highest percentage of variance explained (EF; Greenwood et al. 1985; Eqn 6) where a negative value indicated that the model performed worse than the null model (mean date of flowering or veraison), and a value above zero indicated that the model explained more variance than the null model (with a maximum value of 1); (ii) the root means squared error (RMSE; Eqn 7), which gives the mean error of the prediction in days; (iii) the Akaike Information Criterion (AIC; Burnham and Anderson 2002; Eqn 8), which rates models in terms of parsimony and efficiency, where the lowest value is associated with the best model.

$$EF = 1 - \left(\frac{\sum_{i=1}^{n} (S_i - O_i)^2}{\sum_{i=1}^{n} (O_i - \bar{O})^2} \right) \quad (6)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (S_i - O_i)^2}{n}} \quad (7)$$

Table 2. Statistical analysis of the four tested models for flowering and veraison using the same dataset.

Model:	SW	SW t_0 = 1 January	UniFORC	UniCHILL
Flowering				
EF	0.80	0.75	0.76	0.79
RMSE	5.4	6.1	6.0	5.6
AIC	3481	3740	3709	3559
Veraison				
EF	0.74	0.57	0.72	0.69
RMSE	8.0	10.2	8.2	8.7
AIC	3845	4299	3909	4018

1033 observations for flowering and 925 observations for veraison were used. SW refers to the model Spring Warming; EF is the efficiency of the model; RMSE is the root means squared error; AIC is the Akaike Information Criterion.

$$AIC = n \times \ln\left(\frac{\sum_{i=1}^{n} (S_i - O_i)^2}{n} \right) + 2k \quad (8)$$

where O_i is the observed value, S_i is the simulated value, $\bar{O}$ is the mean observed value of the dataset used, n is the number of observations, k is the number of parameters.

Once a model was selected, the parameter estimates were optimised in order to fix parameters (excluding F^*) to the same values for all varieties and both flowering and veraison stages.

Model validation

The selected model was validated for the most frequently represented varieties (11 in total) for which supplementary independent data had been obtained after model parameterisation. Prior validation, the F^* values of the selected model were fitted for flowering and veraison for each of the 11 varieties from the parameterisation data; these values were validated with the validation data using the statistical criterion of EF and RMSE.

Results

Model selection

The four model types (two versions of SW, UniFORC, UniCHILL) were compared for the same number of observations, i.e. 1033 observations for flowering and 925 observations for veraison. Overall, there was very little difference between UniFORC, UniCHILL and SW in terms of efficiency (Table 2). However, the SW model (all parameters fitted) was more efficient than any of the other models for both flowering and veraison as indicated by the EF and RMSE values. The lower AIC values obtained for the SW model with unfixed parameters for both flowering and veraison indicated that SW was the best model with regard to the trade-off between parsimony and efficiency. SW using the classical parameter value of 1 January as t_0 was the least efficient model overall. The SW model with unfixed parameters for t_0 and Tb was subsequently chosen for our study.

Optimisation of a single model for flowering and veraison

More data from the parameterisation dataset could be used to fit the SW model, which unlike the UniCHILL model does not require temperature data prior to 1 January. The UniCHILL

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Table 3. Efficiency of the Spring Warming model to predict flowering and veraison for different sets of parameter estimates of t_0 and *Tb* (*F** is adjusted for each stage).

Parameters	t_0 (d)	*Tb* (°C)	Flowering	Veraison
t_0 and *Tb* fitted on flowering dates	56	3	0.79	0.69
t_0 and *Tb* fitted on veraison dates	92	4	0.71	0.72
t_0 and *Tb* fixed	60	3	0.79	0.70
t_0 and *Tb* fixed	60	0	0.76	0.72

1092 observations were used for flowering and 980 observations were used for veraison.

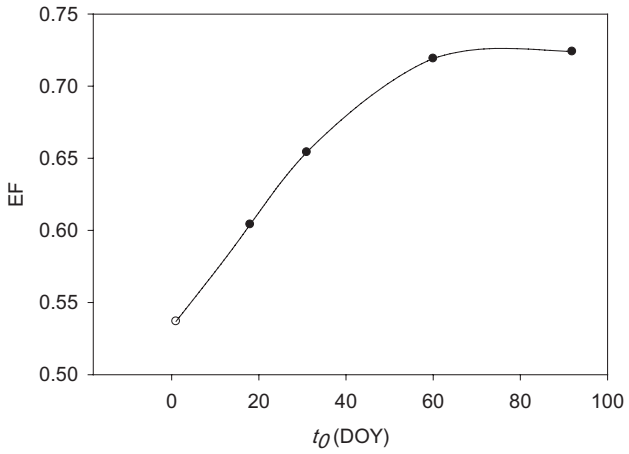


Figure 2. Change in the efficiency (EF) of veraison model (Spring Warming, *Tb* value of 0°C) in response to changes in t_0.

model requires temperature data from 1 September of the prior year as it describes the action of temperature on the dormancy phase. The SW model was fitted again with 1092 flowering observations and 980 veraison observations. The best estimates of parameters t_0 and *Tb* for the SW model fitted on flowering dates were 56.4 days and 2.98°C, respectively (Table 3). For the sake of simplicity for users, these values were rounded to 60 days and 3°C without altering the efficiency of the model for flowering. When applied to veraison, the parameter estimates of 60 days and 3°C slightly decreased the model efficiency (0.70 vs 0.72) compared to the parameter estimates fitted on veraison dates. Conversely, the parameter estimates of t_0 at 92 days and a *Tb* value of 4°C fitted for veraison dates greatly reduced the efficiency of the model for flowering compared to the parameter values obtained for fitting the model to flowering dates (0.79 vs 0.71). Since our aim was to develop one model that could be used for a diverse range of varieties for the timing of appearance of these two phenological stages, we looked for parameter estimates that could therefore optimise the efficiency of the veraison model without unduly altering the efficiency of the model for predicting flowering. Further fitting of veraison (Table 3) by maintaining t_0 at 60 days yielded a *Tb* estimate of 0°C, increasing the efficiency of the veraison model slightly (0.72 vs 0.70) and reducing slightly the efficiency of the flowering model (0.76 compared vs 0.79). A greater range of values for parameters t_0 and *Tb* were then investigated to further confirm the choice of initial parameter estimates that were optimised for the veraison model (t_0 value of 60 days, *Tb* value of 0°C). The efficiency of the model did increase as t_0 increased from 0 to 60 days (*Tb* at 0°C) after which the efficiency remained stable (Figure 2); a decrease

Table 4. Statistical analysis of the optimisation process for *Tb* for veraison using the model Spring Warming t_0 at 60 days.

	Tb (°C)						
	0	**3**	**5**	**7**	**10**	**12**	**15**
EF	0.72	0.70	0.66	0.58	0.26	−0.55	−2.17
RMSE	8.14	8.40	9.00	9.90	13.17	19.09	27.35

EF is the efficiency of the model; RMSE is the root means squared error.

Table 5. Comparison of efficiency and quality of prediction of flowering and veraison between the new Spring Warming model (New SW) parameters (t_0 at 60 days, *Tb* of 0°C) and the classical Spring Warming model (Classical SW) parameters (t_0 at 1 day, *Tb* of 10°C).

	Flowering		Veraison	
	New SW	**Classical SW**	**New SW**	**Classical SW**
t_0	60	1	60	1
Tb	0°C	10°C	0°C	10°C
EF	0.76	0.73	0.72	0.14
RMSE	5.9	6.3	7.7	14.3

1092 observations were used for flowering; 980 observations were used for veraison.

in the efficiency of the model occurred when *Tb* increased from 0°C to 15°C (t_0 at 60 days; Table 4). This confirmed the choice of estimates for parameters t_0 and *Tb* following the optimisation procedure. The model SW using the new parameter estimates of 60 days for t_0 and a *Tb* value of 0°C was slightly more efficient for flowering than the classical model of SW (GDD) with the parameters t_0 at 1 day and a *Tb* value of 10°C. The new model was substantially more efficient for veraison (Table 5).

Gladstones (1992) suggested that phenology predicted by the temperature summation method of GDD was improved when the average temperature was capped at 19°C. That is, for every average daily temperature greater than 19°C, the value of 19 replaces the actual average in the calculation of the thermal summation. We tested this hypothesis for the model SW with t_0 at 60 days and *Tb* fitted, by limiting the maximum average temperature value used in the model between the values of 15°C and 25°C. This resulted in a decreased efficiency of the

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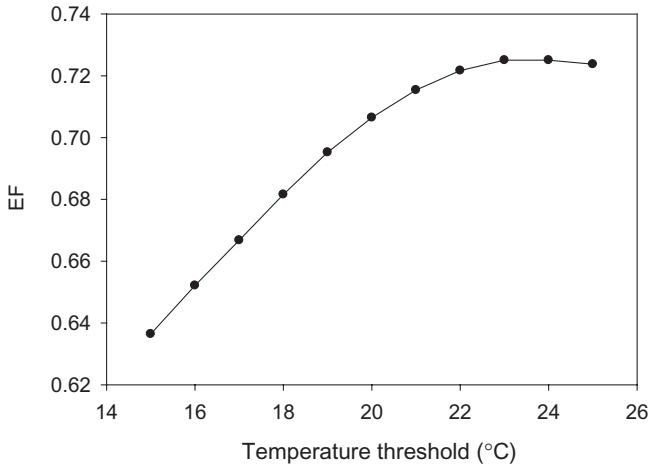


Figure 3. The effect of capping the maximum temperature on model efficiency (EF) for Spring Warming, t_0 at 60 days.

model for any value of the maximum possible average temperature (Figure 3). The suggestion by Gladstone (1992) was therefore not incorporated into our model and daily average temperatures remained uncapped.

Spring Warming with a base temperature (*Tb*) of 0°C calculated from the 60th day (t_0) of the year was therefore selected as the best model. This model is referred to as the Grapevine Flowering Veraison model (GFV) from hereon.

Model validation

For both flowering and veraison, it was observed that the dispersion of data was similar between the data used to parameterise and validate the model on a varietal level (Figures 4,5). However, flowering was, in general, less dispersed for both the model parameterisation and validation compared with veraison. When the model was tested for each variety (EF values in Tables 6,7), in general, the efficiencies were higher than that obtained for the species (the combined cultivar values, EF of 0.76 for flowering, and 0.72 for veraison in Table 3).

For flowering, the efficiencies obtained with the validation dataset for the 11 varieties were always greater than that of the null model (average date). In general, the efficiency was as expected, higher with the parameterisation dataset than with the validation dataset given that the *F** value used for the validation process was estimated from the data used for model parameterisation. However, for most varieties (7 out of 11), the loss of efficiency was less than 20%. Notably, the efficiency of the Chardonnay, Merlot, Grenache and Syrah models was not decreased by more than 10% with the independent validation data compared to the parameterisation data (Table 6). The quality of prediction for each variety (RMSE) remained comparable between both the parameterisation data and the validation data. However, less data were used for the validation process. Regardless, the RMSE values were less than 1 week for all varieties for model parameterisation and validation.

For veraison, four varieties had less than a 20% reduction in efficiency with the validation data (Table 7). For Cabernet franc, Cabernet-Sauvignon, Chardonnay and Riesling, the model still simulated veraison better than the null model (average date). For Chasselas and Pinot noir, negative efficiencies were obtained; however, fewer observations were used corresponding to a reduced spatial and temporal representation for these varieties. Although there were more observations for Ugni blanc than for Chasselas and Pinot noir, a negative EF was obtained for Ugni blanc model validation. This could be attributed to the larger sum of squares (data not shown) between the prediction and observation values for the validation dataset. The quality of prediction (RMSE) was similar for all varieties for the parameterisation data with a maximum value of 7.81 days; the quality was slightly less for the validation data with a maximum value obtained of 9.79 days. However, for most varieties (10 out of 11), there was a difference of less than 3 days between the RMSE for the parameterisation and validation data.

Discussion

Relevance of the GFV model for grapevine

We modelled flowering and veraison using process-based models (Chuine et al. 2003) so far untested for the grapevine with the aim of developing the simplest model possible at the species level. The proposed model GFV with a start date of the 60th DOY (t_0) for application in the Northern Hemisphere; and base temperature, *Tb*, of 0°C showed the best overall performance, representing the best balance between complexity and performance compared to the other possible model choices. The model UniCHILL, which takes into account the effect of chilling temperature during the 'dormancy phase' to break bud rest, was no more efficient or parsimonious, although it may represent a more realistic model in terms of temperature influences during development (García de Cortázar-Atauri et al. 2009). However, for the purpose of a general model, the results show that flowering and veraison can be predicted with good precision based solely on the basis of heat units.

The GFV model proved more efficient than the current classic model of GDD using a base temperature of 10°C from 1 January (in the Northern hemisphere). The base temperature of 10°C has been proposed to represent a threshold above which physiological processes are of importance for phenological development. Our results indicate that either physiological processes influencing phenological development below 10°C could be of more importance than currently thought and/or that the threshold temperature that is optimal for model prediction is not necessarily the temperature threshold for the underlying physiological processes of the developmental stage. The base temperature of 0°C has the advantage for model users in that it is simpler to calculate during the growing season when minimum temperatures are less likely to drop below 0°C. Therefore, in such cases, its application represents a simple addition of accumulated daily average temperatures (from the 60th DOY).

Notably, a recent model successfully adapted to simulate grapevine bud break in cool European regions (Nendel 2010) used the same parameter values for the base temperature and start day for the thermal summation (t_0 at 60 days and *Tb* value of 0°C) that were found optimal in this study. Given that the datasets used and the optimisation processes differed between this study and that of Nendel (2010), this agreement of parameter values indicates a convergence of different phenological stages to the same thermal summation model. Other studies (Williams et al. 1985b, Duchêne et al. 2010) have also proposed initial dates very close to this proposed value of 60 days (20 February and 15 February, respectively). However, to our knowledge, this study is the first to test and confirm the pertinence of this value for a dataset containing such a wide range of varieties, locations and years for flowering and veraison. The parameter values are to be further tested in the future for Southern hemisphere data (where the t_0 is equivalent to the 242nd DOY).

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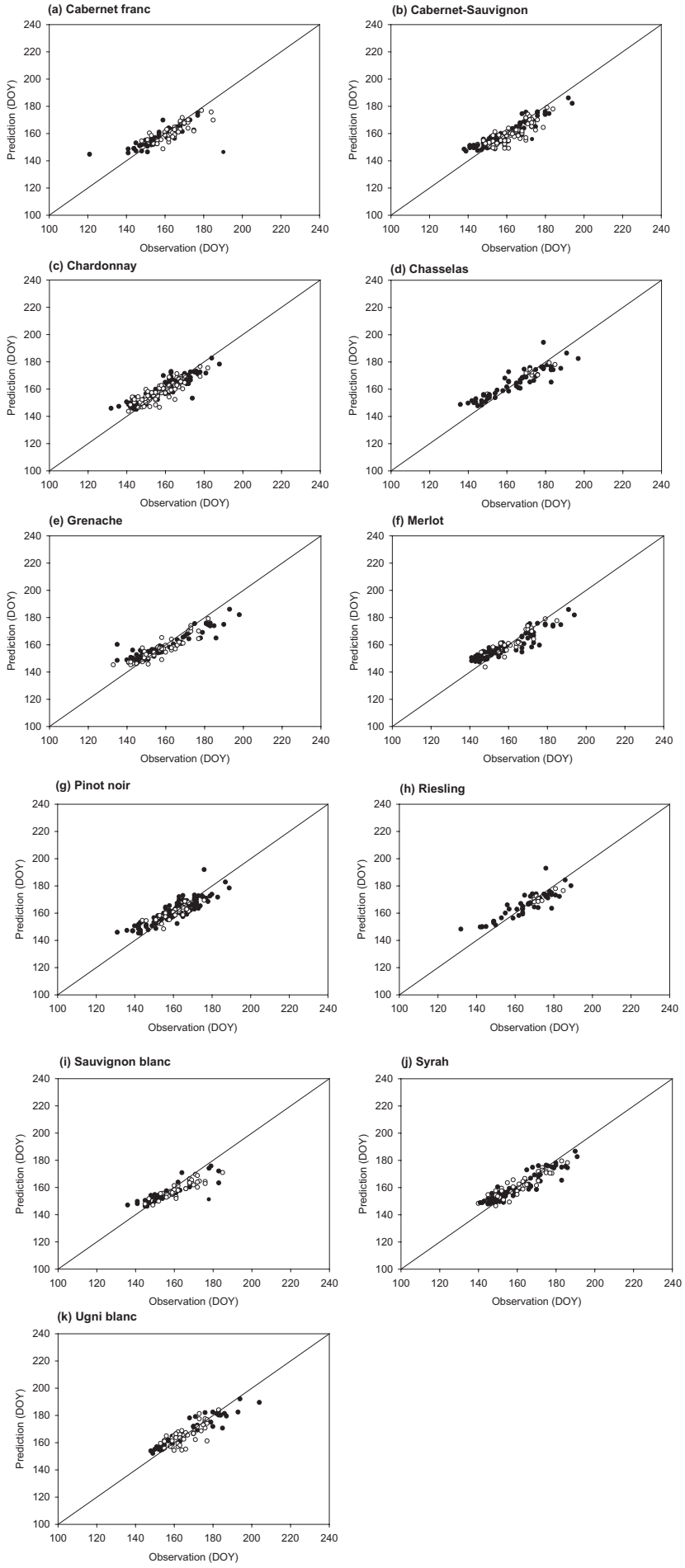


Figure 4. Observed and simulated dates of flowering for 11 varieties using the Grapevine Flowering Veraison model. Closed circles (●) represent data used for the model parameterisation; open circles (○) represent data used for the model validation.

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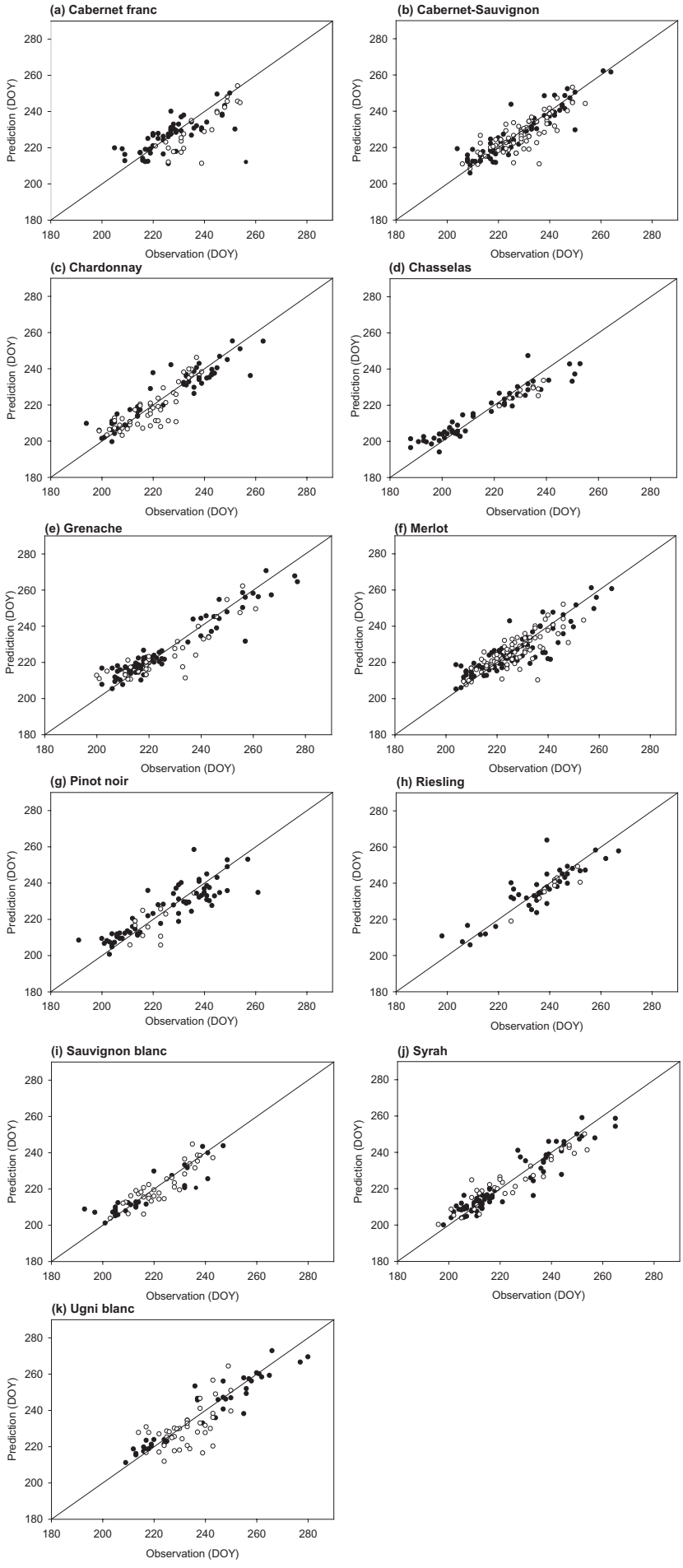


Figure 5. Observed and simulated dates of veraison of 11 varieties using the Grapevine Flowering Veraison model. Closed circles (●) represent data used for the model parameterisation; open circles (○) represent data used for the model validation.

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Table 6. Statistical analysis of the Grapevine Flowering Veraison model parameterised for flowering for the 11 varieties used for the validation process.

Variety	*F**	*n*		EF		RMSE	
		Model parameterisation	Model validation	Model parameterisation	Model validation	Model parameterisation	Model validation
Cabernet franc	1225	57	45	0.73	0.53	5.13	5.45
Cabernet-Sauvignon	1270	70	62	0.83	0.46	3.46	3.65
Chardonnay	1217	100	71	0.78	0.73	5.15	4.69
Chasselas	1274	59	9	0.79	0.31	6.67	3.91
Grenache	1269	92	49	0.78	0.79	4.90	2.75
Merlot	1266	83	24	0.78	0.77	4.24	1.75
Pinot noir	1219	122	29	0.76	0.66	5.50	4.19
Riesling	1242	47	9	0.76	0.45	3.37	0.86
Sauvignon blanc	1238	37	57	0.77	0.63	2.72	3.27
Syrah	1277	92	35	0.84	0.77	3.68	2.33
Ugni blanc	1376	45	50	0.85	0.54	2.81	2.72

*F** is the critical degree-day sum (above 0°C) fitted for each variety that was estimated from the model parameterisation dataset. *n* is the number observations by variety used for the model choice and validation process. EF is the efficiency of the model; RMSE is the root means squared error in days.

Table 7. Statistical analysis of the Grapevine Flowering Veraison model parameterised for veraison for 11 varieties used for the validation process.

Variety	*F**	*n*		EF		RMSE	
		Model parameterisation	Model validation	Model parameterisation	Model validation	Model parameterisation	Model validation
Cabernet franc	2655	57	22	0.62	0.06	6.54	9.79
Cabernet-Sauvignon	2641	66	105	0.83	0.27	5.75	8.03
Chardonnay	2541	54	48	0.86	0.62	6.65	6.32
Chasselas	2342	53	9	0.88	–0.26	6.11	6.04
Grenache	2750	80	36	0.90	0.76	5.67	7.56
Merlot	2627	90	71	0.79	0.62	6.49	6.44
Pinot noir	2507	70	10	0.78	–1.82	7.81	8.07
Riesling	2584	43	9	0.77	0.27	7.06	4.48
Sauvignon blanc	2517	29	33	0.82	0.74	6.17	5.13
Syrah	2598	78	32	0.90	0.84	5.19	6.19
Ugni blanc	2777	41	49	0.90	–0.10	6.32	9.05

*F** is the critical degree-day sum (above 0°C) fitted for each variety that was estimated from the model parameterisation dataset. *n* is the number observations by variety used for the model choice and validation process. EF is the efficiency of the model; RMSE is the root means squared error in days.

Heat requirements at the varietal level: model application

Using the phenological records, we parameterised the GVF model for a large number of varieties (43 varieties for flowering and 45 varieties for veraison corresponding to a total of 1092 and 980 observations, respectively) and the model successfully predicted flowering and veraison for 11 varieties using an independent dataset (440 flowering observations and 424 veraison observations). For both phenological stages, the average quality of prediction by variety (RMSE) was in most cases less than 1 week. The model validation indicated that the model can be very efficient at the variety level, but the predictive power was not necessarily equivalent for all varieties and both phenological stages. The data used for parameterisation and validation correspond to 55 and 67% of all the data collected in the study (Table 1) for flowering and veraison, respectively. Further work will be necessary to refine the *F** values for each variety by combining all records available as presented in the complete dataset summarised in Table 1. However, for the 11 varieties used in the validation process, the initial estimates of *F** should not change substantially given the similarity of results obtained for model selection and validation.

It is possible that varieties may differ in their rate of development between stages and their sensitivity to a given temperature may change as a function of this rate (Pouget 1966, Buttrose 1969, Moncur et al. 1989, Calo et al. 1994). One limitation of the type of model approach presented is that varieties

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will invariably be at different stages during the developmental cycle when the thermal summation begins and their individual rate of development and temperature sensitivity would not be accounted for. Some recent work has considered adapting more complex models to specific varieties (Caffarra and Eccel 2010) and this is an area of research that needs more investigation. However, the GFV model remains advantageous for (i) understanding differences in phenological timing for a wide range of varieties compared within the same modelling framework, (ii) when considering rare or less data-rich varieties for which a separate model may be difficult to achieve, and (iii) its simplicity for the user. Given that the data used were spatially and temporally diverse, corresponding to a wide range of varieties, the current database can be used to explore the GFV model on the varietal level in the future.

Model implications in a context of climate change

Climate change is predicted to advance phenology and ripening; this can be countered to some extent by later ripening clones and some viticultural practices such as late pruning (Friend and Trought 2007 and references within). However, another possibility is to consider changing to different varieties, which could potentially develop and ripen later in the season. This paper has taken the first steps towards successfully predicting flowering and veraison for a range of varieties using one model that in application will help identify suitable varieties for selected climates.

Phenological modelling with climate change scenarios can be used to predict the distribution of varieties in the future (see Duchêne et al. 2010; Garcia de Cortazar-Atauri et al. 2010). The current model was calibrated using a database containing a diverse range of varieties; therefore, this model can be used to better characterise heat requirements of a wide range of varieties and of varieties for which little information is known thus far. In combination with an understanding of future climate change scenarios, such information will allow viticulturists to have a better understanding of which varieties may better perform in future temperature regimes, and direct them in selection of alternate varieties.

Conclusion

We have shown that general process-based models can be successfully applied and validated for the grapevine. A simple model, GFV, corresponding to SW (t_0 at 60 days, *Tb* value of 0°C) has been selected, optimised and shown to be efficient to predict flowering and veraison at the species and varietal level. The model was validated and had greater predictive power than existing models. Its simplicity makes it easy to use, and enables further adoption of the model to predict the varietal timing of flowering and veraison under a changing climate.

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